In this document, there are 1804 words outside of math formulas.

A. Separating the Dynamics from the initialisation of a stochastic process

Fix a domain for stochastic processes $D \subseteq \mathbb{R}^N$ and a time index set $I \subset [0, \infty)$ (so I can be discrete as well, $I = \mathbb{N}$).

On a measurable space $(\Omega, \mathcal{F})$ a **stochastic process** is an I-indexed sequence of random variables with values in D, i.e. a map $X : I \times \Omega \to D$ where $X(t, \cdot)$ is measurable at each fixed time $t \in I$.

This setup already fixes a starting position — the random variable X_0 with its law $\mathbb{P}_{X_0}$. So there are two aspects to this definition that we need to separate conceptually.

Recall that a Markov chain is a stochastic process X satisfying the simple Markov property

$$\forall t_1 < \ldots < t_{k+1} \in I : \quad \mathbb{P}_{X_{t_{k+1}} \mid X_{t_1}, \ldots, X_{t_k}} = \mathbb{P}_{X_{t_{k+1}} \mid X_{t_k}}.$$

A Markov process induces a kernel (*transition matrix* if D is discrete)

$$\text{for any } t \in I, \quad \begin{aligned} &P_t : D \times \mathcal{F} \to [0, 1] \\ &P_t(x, A) := P(X_t \in A \mid X_0 = x), \end{aligned}$$

but a kernel on its own does not determine a stochastic process, not at least until a *starting point* (or distribution) is specified — different processes started at different points can share the same kernel (dynamics). In discrete time and space, it is known that a transition matrix with a choice of a starting point induces a Markov chain.

A stochastic differential equation of the form $dX_t = b(t, X)dt + \sigma(t, X)dw_t, X_0 = x_0$ may be solved by a stochastic process. While the equation + the starting point are what determine the solution, and we are used to saying that the expression $dX_t = b(t, X)dt + \sigma(t, X)dW_t$ is meaningless on its own, it is useful to treat the two as conceptually standalone pieces of the definition — akin to how the Markov kernel is a standalone object paired with each Markov chain. After all, the same b, σ induce different solutions for different starting points x_0.

B. Markov chain action on the space of distributions

C. The sampling problem

As described in [Andrea], we consider 3 classes of sampling problems depending on what is known:

Task: **produce a sample** $x \sim \mu$ where

	we know	but do not know
Case I.	the distribution μ	
Case II.	a family of distributions and a sample sequence $(\mu_\theta)_\theta$ and $(x_i)_i \overset{\perp}{\sim} \mu_\theta$	the parameter θ, and hence which μ_θ
Case III.	a sample sequence only $(x_i)_i \overset{\perp}{\sim} \mu$	the distribution μ

We will work in case I. for now and assume we know the distribution μ. Diffusion based sampling is the approach of approximating μ by a sequence $(\nu_t)_t$

D. The Fokker-Planck equation

While we define[1] a D-valued stochastic process $(Y_t)_t$ on a probability space $(\Omega, \mathbb{P})$ as a map of shape $Y : I \times \Omega \to D$ measurable in the Ω argument, what is *relevant* are all its laws $\mathbb{P}_{Y_t}, t \geq 0$, as well as all its joint laws

$$\text{for all } n \text{ and } t_1, ..., t_n, \quad \mathbb{P}_{(Y_{t_1}, ..., Y_{t_n})} : \mathcal{B}(D^n) \to [0, 1].$$

So from the perspective of time flowing, what is relevant is not so much the sequence $(Y_t)_t$ of the *variables*, but rather the sequence $(\mathbb{P}_{Y_t})_{t\geq 0}$ of their *laws*.

If Y admits probability densities $p(t, x) = \mathrm{d}\mathbb{P}_{Y_t}/dx$, it is natural to consider the sequence of densities $(p(t, \cdot)_t)_t$ and ask if their evolution is driven by a partial differential equation

$$\frac{\partial p(t, x)}{\partial t} = F[t, x, p, \partial_x p, \partial_{xx} p, ...], \text{for some } F,$$

and in particular, **what is that $\boldsymbol{F}$?** In this section, we answer that question positively for Itô diffusions.

D.1. Properly generalizing $\partial_t p(t, x)$

We want to construct a differential equation describing the evolution of the laws of Y_t. Inspired from PDEs on regular functions of the form $\partial_t p(t, x) = F[t, x, \partial_x, ...]$ in mind, we might be tempted to simply replace the function p by the law $\mathbb{P}_{Y_t}$ and develop an equation for $\partial_t \mathbb{P}_{Y_t}$ (if exists). This naïve way leads to inconsistencies that we outline below.

In particular, a well-behaved generalization of $\partial_t p(t, x)$ should be stable under changes to the initial conditions (the law of Y_0). We now show that $\partial_t \mathbb{P}_{Y_t}$ is not.

[1]usually for an interval $I \subseteq \mathbb{R}_{\geq 0}$ and a $\mathcal{B}$orel $D \subseteq \mathbb{R}^N$

D.1.1. The problem with $\partial_t \mathbb{P}_{Y_t}$

> **Context.** For two random variables ξ, η and a real (rate) $r > 0$ construct the processes
>
> $$X_t = \xi + r\,t \quad \text{and} \quad Y_t = \eta + r\,t, \tag{1}$$
>
> satisfying the SDEs
>
> $$\begin{aligned} dX_t &= r\,dt \\ X_0 &= \xi \end{aligned} \quad \text{and} \quad \begin{aligned} dY_t &= r\,dt \\ Y_0 &= \eta \end{aligned}$$
>
> sharing the same dynamics $dX_t = r\,dt = dY_t$ but having different initial conditions ξ, η.

So we'd want our generalization of $\partial_t p$ to be the same on both processes.

Derivation. Evaluate $\partial_t \mathbb{P}|X_t\rangle|_{t=0}$ on $[a, \infty)$ for $a \in \mathbb{R}$:

$$\begin{aligned} \partial_t\, \mathbb{P}(X_t \geq a)|_{t=0} &= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(\mathbb{P}(X_\varepsilon \geq a) - \mathbb{P}(X_0 \geq a)) \\ &\overset{(1)}{=} \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(\mathbb{P}(\xi + r\varepsilon \geq a) - \mathbb{P}(\xi \geq a)) \end{aligned}$$

(express[2] $(\xi + r\varepsilon \geq a) = (\xi \geq a) \cup (a - r\varepsilon \leq \xi < a)$)

$$= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \mathbb{P}(a - r\varepsilon \leq \xi < a), \tag{2}$$

And similarly for Y:

$$\partial_t\, \mathbb{P}(Y_t \in [a, \infty))|_{t=0} = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \mathbb{P}(a - r\varepsilon \leq \eta < a).$$

It is evident now that the derivative depends directly on the laws of ξ and η. If the two were chosen according to, for example, $\xi \sim \mathrm{Unif}(0, 1)$ and $\eta \sim \mathrm{Unif}(0.5, 1)$, and we pick $a = 1$, the limits above would evaluate to

[2] Having the usual notation for events in mind, $(\xi \geq a) \equiv \{\omega \in \Omega \mid \xi(\omega) \geq a\}$

$$\begin{aligned} \partial_t \mathbb{P}(X_t \geq 1) &= r, \\ \partial_t \mathbb{P}(Y_t \geq 1) &= 2r, \end{aligned} \tag{3}$$

because in that case

$$\begin{aligned} &\mathbb{P}(1 - r\varepsilon \leq \xi < 1) = r\varepsilon \\ \text{but} \quad &\mathbb{P}(1 - r\varepsilon \leq \eta < 1) = 2r\varepsilon. \end{aligned} \qquad \square$$

That is,

> **Conclusion.** the time derivative of the laws of X_t, Y_t is not invariant to change of initial conditions.

Why does $\partial_t \mathbb{P}|Z_t\rangle$ vary?

It turns out we can easily visualize this phenomenon if we consider

1. the speed at which (the laws of) X_t and Y_t move through $\mathbb{R}$, and,
2. the speed at which individual sets $A \subseteq \mathbb{R}$ gain or lose weight[3] as t varies, under $\mathbb{P}|X_t\rangle$ and $\mathbb{P}|Y_t\rangle$,

and realize that $\partial_t \mathbb{P}|X_t\rangle$ measures the latter, while the invariant quantity is the former.

In Figure 1, the highlighted areas indicate the weight gained by $[1, \infty)$ as time progresses from 0 on. While both X_t and Y_t, as well as their plotted

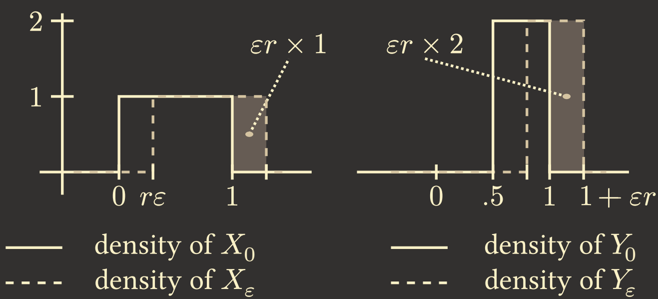


Figure 1: The densities of X and Y at times $t = 0$ and $t = \varepsilon$. Here again $\xi \sim \text{Unif}(0, 1)$ and $\eta \sim \text{Unif}(0.5, 1)$.

[3] measure

densities, are moving to the right at the same rate r (both driven by a shared $dX = r\,dt = dY$), the rate of weight gain per unit time of $[1, \infty)$ is twice as large in the Y case as in the X case.

D.1.2. The time-derivative of $\mathbb{P}|X_t\rangle$

> **Context.** Fix a process $Z_t = \zeta + r\,t$ where ζ is a random variable with CDF F_ζ and continuous density $p_\zeta = F'_\zeta$.

Derivation. Pick $a \in \mathbb{R}$ and evaluate the probability gain over $(-\infty, a)$ at t:

$$\begin{aligned} \mathbb{P}(Z_t \le a) - \mathbb{P}(Z_0 \le a) &= \mathbb{P}(\zeta + r\,t \le a) - \mathbb{P}(\zeta \le a) \\ &= \mathbb{P}(\zeta \le a - r\,t) - \mathbb{P}(\zeta \le a) \\ &= -\mathbb{P}(a - r\,t \le \zeta \le a) \\ &= -\int_{a-r\,t}^{a} p_\zeta(a) da \end{aligned} \tag{4}$$

(by the mean value theorem, for some $a' \in (a - r\,t, a)$)

$$= -r\,t \times p_\zeta(a').$$

This gives us a Taylor expansion arount $t = 0$:

$$\mathbb{P}(Z_\varepsilon \le a) = \mathbb{P}(Z_0 \le a) - r\varepsilon \times p_\zeta(a) + o(\varepsilon^2). \tag{5}$$

The term $r\varepsilon \times p_\zeta(a)$ corresponds to the highlighted rectangles in Figure 1, and where the invariant quantity r is "corrupted" by the multiplier $p_\zeta(a) = F'_\zeta(a)$.

Recall that we can recover the law of a real random variable from its CDF $F(a) = \mu(-\infty, a]$, by the Lebesgue-Stieltjes integral $\mu(A) = \int_A dF(a)$. Denote[4] the measure recovered this way by $\mathbb{P}|F\rangle$. Over all $a \in \mathbb{R}$, Equation 5 is an equation of CDFs, so it naturally induces an equation of *laws*.

[4]No ambiguity can arise with the notation $\mathbb{P}|\zeta\rangle$ and $\mathbb{P}|F\rangle$ because ζ is of shape $\Omega \to \mathbb{R}^N$, while F is of shape $\mathbb{R} \to \mathbb{R}$. And we have $\mathbb{P}|\zeta\rangle = \mathbb{P}\big|F_\zeta\big\rangle$ if ζ is $\mathbb{R}$-valued.

The (signed) measure induced by the first order term $rp_\zeta(a) = rF'_\zeta(a)$ is then $r\mathbb{P}\big|F'_\zeta\big\rangle$, so Equation 5 in measure speak reads

$$\mathbb{P}|Z_\varepsilon\rangle = \mathbb{P}|Z_0\rangle - \varepsilon r\,\mathbb{P}\big|F'_\zeta\big\rangle + o(\varepsilon^2).$$

To recover r when taking the derivative $\partial_t\mathbb{P}|Z_t\rangle$ now it is natural to also divide by $\mathbb{P}\big|F'_\zeta\big\rangle$ in Radon-Nikodym sense:

$$r = \frac{\mathrm{d}\partial_t\mathbb{P}|Z_t\rangle}{\mathrm{d}\mathbb{P}\big|F'_\zeta\big\rangle},$$

or, writing it in higher-order derivative notation,

$$r = \left.\frac{\partial\,\mathrm{d}\mathbb{P}|Z_t\rangle}{\partial t\,\mathrm{d}\mathbb{P}\big|F'_\zeta\big\rangle}\right|_{t=0},$$

which indicates how from the increment of $\mathbb{P}|Z_t\rangle$ we cancel a time t and a density $\mathbb{P}\big|F'_\zeta\big\rangle$ factor. □

That is, we reach the following

Conclusion. A natural initial-distribution-invariant quantity based on the time derivative of Z_t's law is its Radon-Nikodym derivative against the signed measure induced by (the Lebesgue-Stieltjes integral with respect to) Z_0's density.

Two problems arise in the derivation of the last expression: for it to work,

- $Z_0 = \zeta$ must admit a (continuous) density $p_\zeta = F'_\zeta$;
- even then, the absolute continuity condition $\mathbb{P}\big|F'_\zeta\big\rangle \ll \mathbb{P}|Y_t\rangle$ must hold.

Thus it is important to go back to Equation 5 and proceed in a way that does not require canceling a F'_ζ term.

D.1.3. Getting out into the box

In Figure 1 we treated the increment $\mathbb{P}|Z_\varepsilon\rangle - \mathbb{P}|Z_0\rangle$ to the first order in ε as a rectangle. We then obtained its width by dividing its mass by its height: $(\varepsilon r \times p_\zeta)/p_\zeta$. Now, while ζ's density p_ζ might not exist, its mass always does.

And if we write $p_\zeta = 1 \times p_\zeta$, the quantity is numerically identical but probabilistically understandable as a unit weight of ζ, which we always can compute from $\mathbb{P}|\zeta\rangle$. Can we divide the area $(\varepsilon r \times p_\zeta)$ by $1 \times p_\zeta$ — a unit mass of ζ then, and recover r?

This would correspond to dividing the mass $(\varepsilon r \times p_\zeta)$ by the mass $(1 \times p_\zeta)$, thus arriving at a *dimensionless* quantity εr. But that is nonsensical, because εr has units of *space*:

$$Z_\varepsilon = \zeta + r\varepsilon,$$

$$\left.\begin{aligned} [Z_\varepsilon] &= \text{space} \\ [\varepsilon] &= \text{time} \end{aligned}\right\} \text{ hence } \begin{aligned} [r] &= \text{velocity, and} \\ [\varepsilon r] &= \text{space.} \end{aligned}$$

That is, the "modified" division

$$\varepsilon r = \frac{\varepsilon r \times p_\zeta}{1 \times p_\zeta} \qquad \left[= \frac{\text{mass}}{\text{mass}}\right]$$

loses a dimension of space. This suggests to add one back, to the numerator. Conceptually this means that instead of as mass over 1D space $(\varepsilon r \times p_\zeta)$, we consider it as a box $(1 \times \varepsilon r \times p_\zeta)$ — a mass over 2D space. Having one extra spatial dimension in the domain of the law amounts to including the $Z_0 = \zeta$ variable and considering the *joint* law

$$\mathbb{P}|Z_\varepsilon, Z_0\rangle : \mathcal{B}(\mathbb{R}^2) \to [0, 1].$$

Let us see now how the time increment of $\mathbb{P}|Z_t, Z_0\rangle$ looks like with the last derivation in mind.

D.1.4. Diffusions

> **Context.** Consider the diffusion process
>
> $$Z_t = \zeta + rt + W_t$$

> for a constant $r > 0$ and an initial $Z_0 = \zeta$ independent from the Wiener process $(W_t)_t$.

Derivation. Similarly to the diffusion-free context, for any $a \in \mathbb{R}$ we have

$$\begin{aligned}&\mathbb{P}(Z_t \leq a) - \mathbb{P}(Z_0 \leq a) \\ &\qquad = -\mathbb{P}(a - r\,t - W_t \leq \zeta \leq a).\end{aligned}$$

In Equation 4 we expressed the probability as the length of the interval times the density of ζ at a point in it, but in this case, the interval is random — its length depends on W_t. We can still evaluate the joint law, though:

$$\begin{aligned}\mathrm{d}\mathbb{P}|Z_0 = x, Z_\varepsilon = z\rangle &= \mathrm{d}\mathbb{P}|\zeta = x, \zeta + r\,t + W_t = z\rangle \\ &= \mathrm{d}\mathbb{P}|\zeta = x, W_t = z - r\,t - x\rangle \\ &\text{(by independence } \zeta \perp W_t)) \\ &= \mathrm{d}\mathbb{P}|\zeta = x\rangle \cdot \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle\end{aligned}$$

That is, the law $\mathbb{P}|\zeta\rangle$ naturally falls of from $\mathbb{P}|Z_0, Z_\varepsilon\rangle$, and we cancel ζ's effect completely via the Radon-Nikodym derivative:

$$\frac{\mathrm{d}\mathbb{P}|Z_0, Z_\varepsilon = y\rangle}{\mathrm{d}\mathbb{P}|Z_0\rangle}(x) = \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle.$$

On sets $(A, B) \in \mathcal{B}(\mathbb{R}^2)$, the join law decomposition above acts by

$$\begin{aligned}\mathbb{P}|Z_0, Z_\varepsilon\rangle(A, B) &= \iint_{(A,B)} \mathrm{d}\mathbb{P}|Z_0, Z_\varepsilon\rangle \\ &= \int_A \left(\int_B \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle \right) \mathrm{d}\mathbb{P}|\zeta = x\rangle \\ &= \iint \underbrace{1_{A,B}(x, x + r\,t + y)\, \mathrm{d}\mathbb{P}|W_t = y\rangle}_{\text{pure evolution}} \mathrm{d}\mathbb{P}|\zeta = x\rangle,\end{aligned}$$

where we see the evolution dynamics integrated along the initial condition ζ. The Radon-Nikodym derivative above then corresponds to disintegrating this expression, i.e. taking the integrand only, which is a map

$$\begin{aligned} \mathbb{R} \times \mathcal{B}(\mathbb{R}) &\to \mathbb{R} \\ (x, B) &\mapsto \int_B 1_{A,B}(x, x + r\, t + y)\, \mathrm{d}\mathbb{P}|W_t = y\rangle \end{aligned}$$

shaped precisely as a Markov kernel. □

Conclusion. While the law of Z_t contains the pure evolution increments in a non-trivial way, extractable (in the $Z_t = \zeta + r\, t$ case) by a Radon-Nikodym derivative against the initial condition's spatial derivative, the joint law of (Z_0, Z_t) contains the entire Z_0 dependence as a clean factor that is easily cancellable by a Radon-Nikodym derivative against $\mathbb{P}|Z_0\rangle$, resulting in

$$\frac{\mathrm{d}\mathbb{P}|Z_t, Z_0\rangle}{\mathrm{d}\mathbb{P}|Z_0\rangle} = \mathbb{P}|Z_t \mid Z_0\rangle.$$

D.2. By Kramer-Moyal expansion

Now that we understand that the proper object to study the distribuitional evolution of an Itô stochastic process is not $\mathbb{P}|Y_t\rangle$ but $\mathbb{P}|Y_t \mid Y_0\rangle$, we present the first approach to deriving the Fokker-Planck equation.

We will derive it by Tailor-expansion of the transition densities, following the general approach in [Walter] or the referenced inside [Garcia-Palacios], sec. 4 and 5. However, we rework it in the language of measures and Markov kernels so that the probability distributions are not assumed to admit densities (against the Lebesgue measure, as done in these resources). This is why we will arrive at a PDE of distributions and not of ordinary functions.

References on this page: `("Walter", "Garcia-Palacios")`

Recall that the Dirac-delta function is the characteristic function with flipped arguments: for all measurable $A \subseteq \mathbb{R}$ and all $y \in \mathbb{R}$, $1_A(y) = \delta_y(A)$. Evaluate that at $y := Y_{t+\varepsilon}$:

$$1_A(Y_{t+\varepsilon}) = \delta_{Y_{t+\varepsilon}}(A).$$

We start from

$$\mathbb{P}[Y_{t+\varepsilon} \mid Y_t] = \mathbb{E}\left[\delta_{Y_{t+\varepsilon}} \;\middle|\; Y_t = y\right].$$

We want to Taylor-expand $\delta_{Y_{t+\varepsilon}} = \delta_{Y_t + (Y_{t+\varepsilon} - Y_t)}$ around Y_t We aim to apply $\frac{d}{d\varepsilon}|_{\varepsilon=0}$.

$$\begin{aligned}\partial_t p_t(y) &= -\partial_y\left(a^{(1)}(y,t)p_t(y)\right) \\ &\quad + \frac{1}{2}\partial_y^2\left(a^{(2)}(y,t)p_t(y)\right)\end{aligned}$$

D.3. By Generators

In infinitesimal form this requires a bit more machinery. We define the infinitesimal generator of X by

$$\begin{aligned}\mathscr{L}\varphi &= \left(x \mapsto \lim_{h \to 0} \frac{P_h\varphi - \varphi}{h}(x)\right) \\ &= \left.\frac{dP_t}{dt}\right|_{t=0}(\varphi)\end{aligned}$$

which tells us by how much a test function φ (thought as a distribution density) is pushed by X_t in an infinitesimal time. For a time-homogeneous $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ this is[5]

$$\mathscr{L} = b^i\partial_i + \frac{1}{2}a^{ij}\partial_i\partial_j, \quad a := \sigma\sigma^\top.$$

Then $u(t,x) := (P_t\varphi)(x)$ satisfies $\partial_t u = \mathscr{L}u$

Next we define the adjoint $\mathscr{L}^*$ and forward Kolmogorov is

[5]using Einstein summation

$$\partial_t p_t(x,y) = \mathscr{L}_y^+ p_t(x,y)$$
$$\mathscr{L}^+\varphi = \frac{1}{2}\partial_i\partial_j a^{ij}\varphi - \partial_i b^i \cdot \varphi$$

where $p_t(x,y)$ is transition density from x to y in time t (for time-homogeneous systems it doesn't matter at which time t_0 we start, only the time difference t).

Then if

$$\rho(x) = \frac{d\mathbb{P}_{X_0}}{dx}(x),$$
$$\rho(t,y) := \int p_t(x,y)\rho(x)\,\mathrm{d}x,$$

Fokker-Planck is

$$\partial_t \rho(t,y) = \mathscr{L}_y^+ \rho(t,y).$$

E. Ornstein-Uhlenbeck

Let $x \sim \mu$ such that $x \perp W$.

$$dZ_t = -Z_t dt + \sqrt{2} dW_t$$
$$Z_0 = x$$

From $e^{-t} de^t Z_t = dZ + Z dt$ we find the solution is

$$\begin{aligned} Z_t &= e^{-t}x + \sqrt{2}e^{-t}\int_0^t e^s\,\mathrm{d}W_t \\ &= e^{-t}x + \sqrt{1-e^{-2t}}\,G_t, \quad (x, G_t) \sim \mu \otimes \mathcal{N}(0,1) \end{aligned}$$

where we rescaled the stochastic integral to

$$G_t := (1/2(e^{2t}-1))^{-1/2}\int_0^t e^s\,\mathrm{d}W_s$$

to get a unit normal[6] $G_t \sim \mathcal{N}(0,1)$.

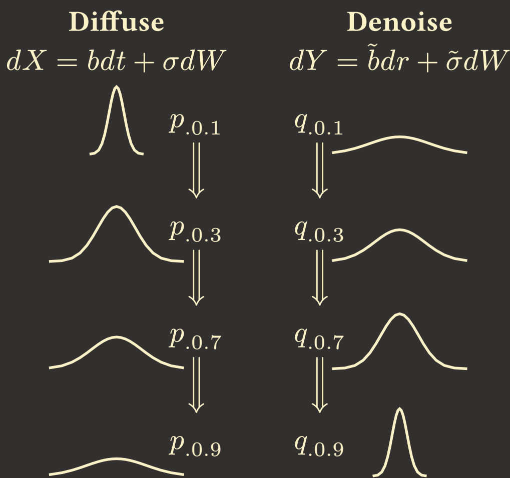
[6] $\int_0^t e^s \,\mathrm{d}W_s$ is normal of variance $V = \int_0^t e^{2s} \,\mathrm{d}s = \frac{1}{2}(e^{2t} - 1)$, thus $G_t := V^{-1/2} \int_0^t e^s \,\mathrm{d}W_s$ is normal of variance 1.

F. The $\nabla \log p$ term

Pick an SDE

$$dX_t = b_t(X_t)dt + \sigma_t dW_t, \quad X_0 \sim \mu$$

and put its distributions $p_t dx = d\mathbb{P}_{X_t}$.



Now let us find $\tilde{b}, \tilde{\sigma}$ that make for a process Y_r having the densities of X_t in reverse order.

So we are after $\tilde{b}, \tilde{\sigma}$ such that $dY_r = \tilde{b}_r(Y_r)dr + \tilde{\sigma}_r dW_r$ has densities $d\mathbb{P}_{Y_r} = d\mathbb{P}_{X_t}$.

Let $q_r dx = d\mathbb{P}_{Y_r}$. Then we desire $q_r = p_t$ for all $r \sim t$.

Both p_t, q_r satisfy the respective Fokker-Plancks

$$\begin{aligned} \dot{p_t} &= F[b_t, \sigma_t, p_t] && \text{for all } t \in I \\ \dot{q_r} &= F\left[\tilde{b}_r, \tilde{\sigma}_r, q_r\right] && \text{for all } r \in I \end{aligned}$$

To connect the $\tilde{b}, \tilde{\sigma}$ to b, σ we need to couple the two equations. Clearly we can get a relationship between $\dot{p}$ and $\dot{q}$ if we ∂_r the equality $q_r = p_t$ (considering $r = r(t)$, we have $\dot{r}$):

$$\dot{q}_r = \partial_r q_r \overset{!}{=} \partial_r p_t = \dot{p}_t \partial_r t = \dot{t}\dot{p}_r.$$

Naturally, if we pick $r \sim t$ to be $r + t = T$, the relation is $\dot{q}_r = -\dot{p}_t$, meaning that p_t flows oppositely to q_t. Applying Fokker-Planck now gives

$$F\left[\tilde{b}_r, \tilde{\sigma}_r, q_r\right] = \dot{t}F[b_t, \sigma_t, p_t].$$

$$\nabla \cdot \left[\tilde{b}_r q_r\right] - \frac{1}{2}\Delta\left[\tilde{\sigma}_r^2 q_r\right] = \dot{t}\left(\nabla \cdot [b_t p_t] - \frac{1}{2}\Delta\left[\sigma_t^2 p_t\right]\right)$$

We have two terms of shape $\nabla \cdot (p_t\ \cdot)$ and two of shape $\Delta(p_t\ \cdot)$; rearrange

$$\nabla \cdot \left[p_t\left(\tilde{b}_r - \dot{t}b_t\right)\right] = \frac{1}{2}\Delta\left[p_t\left(\tilde{\sigma}_r^2 - \dot{t}\sigma_t^2\right)\right]$$

By recalling that $\Delta = \operatorname{div} \circ \operatorname{grad}$, pull a div from both sides:

$$\operatorname{div}\left[p_t\left(\tilde{b}_r - \dot{t}b_t\right)\right] = \operatorname{div}\left[\frac{1}{2}\operatorname{grad}\left[p_t\left(\tilde{\sigma}_r^2 - \dot{t}\sigma_t^2\right)\right]\right].$$

Put $s_t \equiv \tilde{\sigma}_r^2 - \dot{t}\sigma_t^2$ for short. Split cases $p_t = 0$ and $p_t > 0$ in the right hand side to pull out a p_t term

$$\nabla[p_t s_t] = \left(1_{p_t>0} + 1_{p_t=0}\right)\nabla[s_t p_t]$$

Assuming we pick $s_t > 0$, note that on $p_t = 0$ we have[7] $\nabla s_t p_t = 0$, so the $1_{p_t=0}$ term vanishes:

$$\begin{aligned} &= p_t \frac{\nabla s_t p_t}{p_t} 1_{p_t>0} \\ &= p_t s_t 1_{p_t>0} \nabla[\ln s_t p_t] \end{aligned}$$

The equation has one obvious solution for $\tilde{b}_r$ now if we drop the div and divide by p_t:

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[7] p_t is non-negative, so at $x \in D$ with $p_t(x) = 0$, the gradient must be zero, otherwise $p_t(x - \varepsilon\nabla p_t(x)) < 0$ for sufficiently small $\varepsilon > 0$

$$\tilde{b}_r = \dot{t} b_t + \frac{1}{2} s_t 1_{p_t>0} \nabla[\ln p_t s_t] \quad \text{for all } r \sim t.$$

For a saner appearance, pick $\tilde{\sigma}_r = \sigma_t$ and $r + t = T$ so that $\dot{t} = -1$ and $s_t = 2\sigma_t^2$, then

$$\tilde{b}_r = -b_t + \sigma_t^2 1_{p_t>0} \nabla[\ln \sigma_t^2 p_t].$$

This derivation works if σ_t is given strictly positive for all times everywhere. If σ_t hits zero, the world shatters.

If $\sigma_t(x) = \sigma_t$ is constant in space, then[8] $\nabla[\ln \sigma_t^2 p_t] = \nabla[\ln p_t]$ and

$$\tilde{b}_r = -b_t + \sigma_t^2 \nabla[\ln p_t] 1_{p_t>0}.$$

[8] $\nabla[\ln \sigma_t^2 p_t] = \nabla[\ln \sigma_t^2 + \ln p_t] = \cancel{\nabla[\ln \sigma_t^2]} + \nabla[\ln p_t] = \nabla[\ln p_t]$